Round 5: the cofactor penalty is self-inflicted

research/isogeny_conductor_gap_round5_20261012. E6 2026-10-12 (stage diagnostic), E7 2026-10-18 (whole-method against matched rho, and relation-phase engineering). Protocol frozen before E6; E7 section added after E6, before E7 ran.

Findings

1. **Round 3 overstated its quotient gain; the real lever is yield (E6).** At matched columns the direct and quotient systems are the same size. A base pushed forward to E/H keeps columns and degree but lands in an index-|H| subgroup: decompositions per target grew as $|H|^{1.001 \pm 0.036}$ for |H| = 2 to 200 (112 arms, 15 curves, 0 lift failures). F4 cost is unchanged on unsolvable targets; solvable ones cost 3–10× more.
2. **Any base inside the targets' subgroup gets that yield (E7a).** A cofactor-cleared base ([h]P for small-abscissa points) costs 0.705× (h = 4) and 0.275× (h = 72) the best fiber-closed table of rounds 1–2, as predicted ($h^{-1/3}$ = 0.63, 0.24). Best cell 7.3× rho (h = 72, 22 bits).
3. **Rho stays ahead.** Slope of $\log_2$ of IC/rho on $\log_2$ #E: 0.122 [0.081, 0.163] (h = 4) and 0.068 [0.022, 0.114] (h = 72): positive, below the derived 1/6.
4. **Binary trace-zero bases (E7b, a known lever ported to ecbench).** Yield 1.94× [1.77, 2.14] over F_2^{23} and 1.48× [1.37, 1.60] over F_2^{19}, with F4 work per relation 1.39× better and 0.87× (worse), since each solvable system costs more.

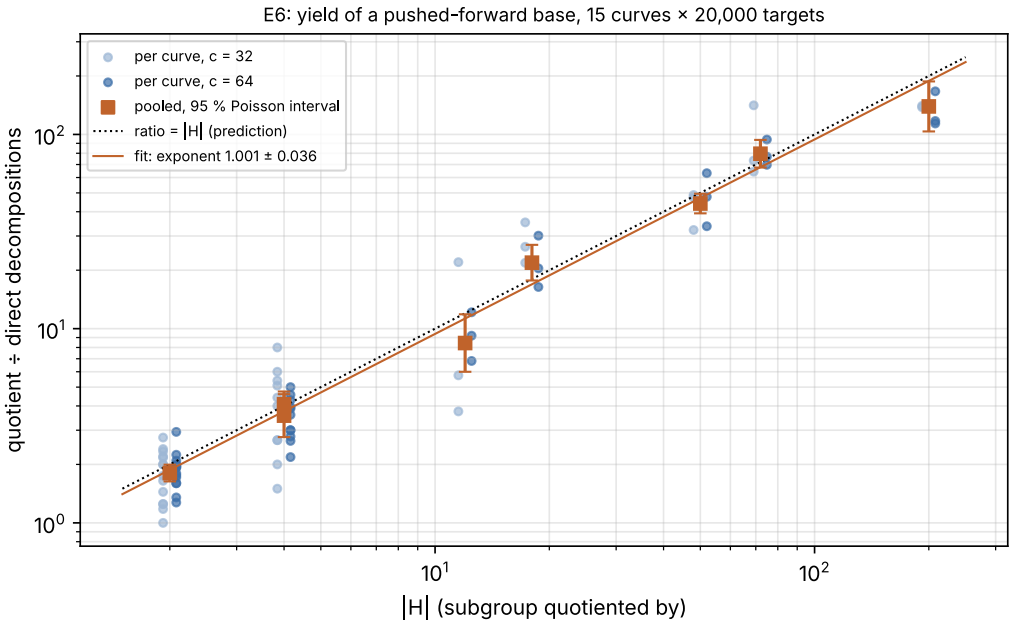


Figure 1: E6: decompositions on E/H ÷ on E, same targets and columns, against |H|.

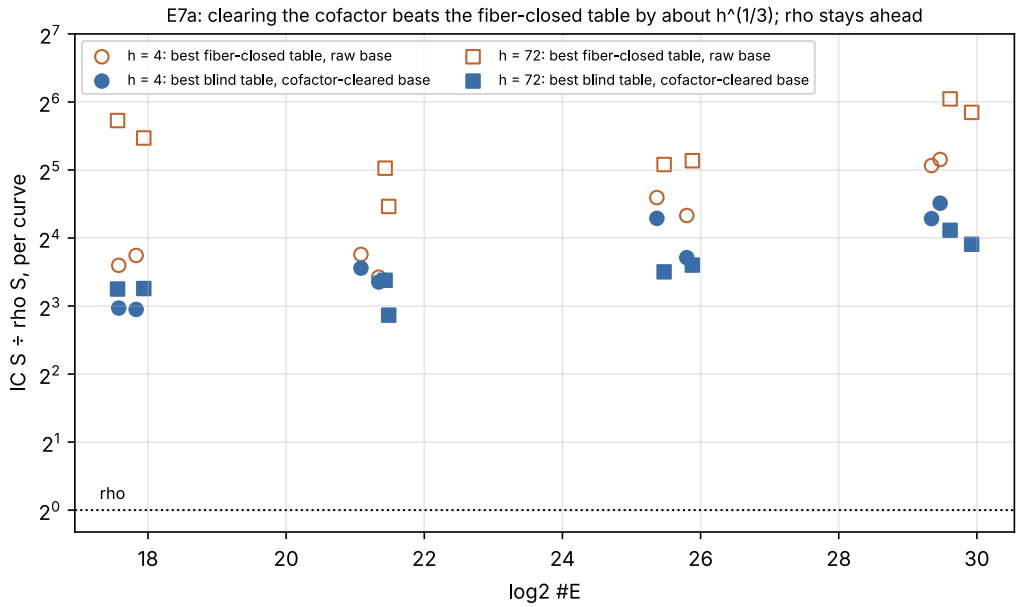


Figure 2: E7a: best IC ÷ rho per curve, fiber-closed table on the raw base (hollow) against blind table on the cofactor-cleared base (filled).

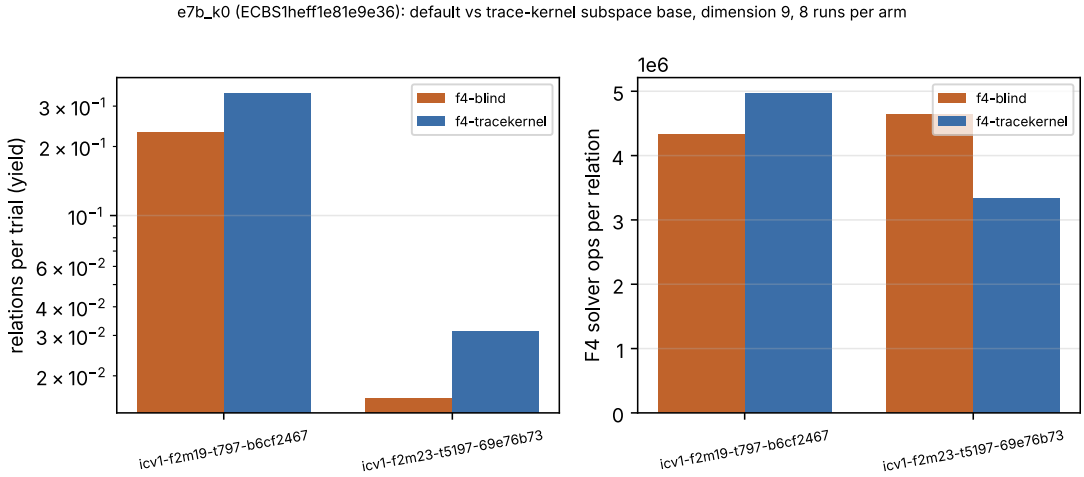


Figure 3: E7b: yield and F4 work per relation, default against trace-kernel base.

Sessions

session	records	audit	isolation earned
e7a_h72 (ECBS1hd38ca558d7e4)	432	ok, 12 replays	L2 380, L1 52
e7a_h4 (ECBS1h54f0529e1598)	480	ok, 12 replays	L2 395, L1 85
e7b_k0 (ECBS1heff1e81e9e36)	80	ok, 12 replays	L2 44, L1 36

E6 records: e6/records.jsonl (not an ecbench session; exact counts). Run reports under reports/.

§5a pricing band (weights 0, 0.1, 1 GAE per op)

- E7a: cleared base cheaper than the best fiber-closed arm on every curve at every weight (ratio 0.18 to 0.97). Cleared over rho spans 7.1 to 27; best cell 7.06 to 7.29 times rho. Rho is cheaper than every IC arm at every weight.
- E7b: trace-kernel MITM beats default MITM on both curves at every weight. Trace-kernel F4 beats default F4 over F_2^{23} over F_2^{19} the pair is unresolved. Priced F4 arms trail rho by 10^4 to 10^6.

Leads, ranked

1. Solvable systems cost 3–10× more to solve; every yield lever moves work there. Measure where (basis completion or solution extraction) and decide consistency first.
2. Affine trace-kernel subspaces for Tr(a) = 1 (a = 1 Koblitz, h = 2) need an offset in the descent.

3. Map where the trace-kernel lever pays (n = 17–29, dimension 8–10) and measure the default base's 2E fraction.
4. Close the prime-field pair-table line: a generic method at 7–17× rho whose gap widens.